1. a) $R_t = \dfrac{[P_t - P_{t-1}]}{(P_{t-1})}$ $\Rightarrow R_1 = \dfrac{P_1 - P_0}{P_0} = \dfrac{8}{100} = 8\%$ $\Rightarrow R_3 = \dfrac{P_3 - P_2}{P_2} = \dfrac{12}{103} = 11.65\%$

$\Rightarrow R_2 = \dfrac{P_2 - P_1}{P_1} = \dfrac{-5}{108} = -4.63\%$ $\Rightarrow R_4 = \dfrac{-5}{115} = -4.35\%$

b) log Returns

$r_t = \ln\left(P_t / P_{t-1}\right)$ $\Rightarrow r_1 = \ln\left(\dfrac{P_1}{P_0}\right)$, $r_2 = \ln\left(\dfrac{P_2}{P_1}\right)$, $r_3 = \ln\left(\dfrac{P_3}{P_2}\right)$

$r_4 = \ln\left(P_4 / P_3\right)$

c) Small returns $r_t = R_t$

$\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \cdots$

$\ln(1-x) = -x - \dfrac{x^2}{2} - \dfrac{x^3}{3} - \cdots$

$\ln\left(\dfrac{1+x}{1-x}\right) = 2\left(x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \cdots\right)$

~~Absolute difference~~ $= |R_t - r_t|$

where R_t - Simple returns

r_t - log returns.

2. $E[R_1] = 15\%$ — $w_1 = 0.4$

$E[R_2] = 7\%$ — $w_2 = 0.6$

Then $E[R_p] = w_1 \cdot E[R_1] + w_2 \cdot E[R_2]$

$= 0.4 \times 15 + 0.6 \times 7$

$= 10.2\%$

If R_1 & R_2 are correlated.
it affects the variance
but $E[R_p]$ remains intact.

3. $var(x+y) = var(x) + var(y) + 2\,cov.(x,y)$.

$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$.

$\because var(ax) = a^2 var(x)$

$R_p = w_1R_1 + w_2R_2$

$var(w_1R_1 + w_2R_2) = var(w_1R_1) + var(w_2R_2) + 2\,cov(w_1R_1, w_2R_2)$

$= w_1^2\,var(R_1) + w_2^2\,var(R_2) + 2\,cov(w_1R_1, w_2R_2)$

$= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,cov(R_1, R_2)$.

$cov(R_1, R_2) / \sigma_1 \cdot \sigma_2 = \rho_{12}$ → correlation coeff.

PAGE NO.:
DATE: / /

4. (A) $\mu = 18\%$ $\sigma = 25\%$

(B) $\mu = 8\%$ $\sigma = 12\%$

Portfolio weighs $w_A = 0.5$ $w_B = 0.5$

a) $\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A \cdot w_B \cdot \rho_{AB} \cdot \sigma_A \cdot \sigma_B$

For $\rho > 0$ it means +ve correlation larger values.

$\Rightarrow \rho = +1$

$= (0.5)^2 (0.25)^2 + (0.8)^2 (0.12)^2 + 2 \times 0.5 \times 0.5 \times 1 \times 0.25 \times 0.12$

$= 0.034225$

$\rho = +1 \Rightarrow \sigma_p = \sqrt{0.03425} = 0.185 = 18.5\%$

$\rho = 0 \Rightarrow \sigma_p = \sqrt{0.019225} = 0.13865 = 13.87\%$

$\rho = -1 \Rightarrow \sigma_p = \sqrt{0.004225} = 0.065 = 6.5\%$

when $\rho = +1$.

$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \cdot 1 \cdot \sigma_A \cdot \sigma_B$

$\sigma_p^2 = (w_A \sigma_A + w_B \sigma_B)^2$ +ve

$\sigma_p = w_A \sigma_A + w_B \sigma_B$ — linear expression.

if $\rho = -1$

Then, $\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 - 2 w_A w_B \sigma_A \sigma_B$

$= (w_A \sigma_A - w_B \sigma_B)^2$ $\quad \sigma_{p_0} = | w_A \sigma_A - w_B \sigma_B |$

Thus, $\boxed{\sigma_{p_0} < \sigma_p}$

$w_A^* (\sigma_A + \sigma_B) = \sigma_B$

$\Rightarrow w_A^* = \sigma_B / (\sigma_A + \sigma_B)$.

5. when $\rho < 1$.

thus, $\sigma_p^2 = w_A^2 \sigma_A^2 + w_B \sigma_B^2 - 2 w_A w_B \sigma_A \sigma_B < (w_A \sigma_A + w_B \sigma_B)^2$.

6.) $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$

$\rho_{12} = 0.4$

$\rho_{13} = 0.1$

$\rho_{23} = 0.2$

$\rho_{ii} = 1 \ \forall i$

$\Sigma_{11} = \rho_{11} \sigma_1 \sigma_1 = 1 \cdot \sigma_1^2$

$\Sigma_{12} = \rho_{12} \sigma_1 \sigma_2 = 0.4 \times 0.25 \times 0.12$

$\Sigma_{13} = \rho_{13} \sigma_1 \sigma_3 = 0.1 \times 0.25 \times 0.04$

$\Sigma_{21} = \rho_{21} \sigma_2 \sigma_1$

$\Sigma_{22} = \rho_{22} \sigma_2^2$

$\Sigma_{23} = \rho_{23} \sigma_2 \cdot \sigma_3$

$$\begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{bmatrix}$$

= 3x3 covariance matrix

b) Σ to be symmetric $\Sigma = \Sigma^T$ of the matrix

$\Sigma_{13} = \Sigma_{31}$ — As these values match diagonal values ~~mathematically~~.

$\Sigma_{12} = \Sigma_{21}$

7. Given, $R_f = 4\%$ Sharpe ratio $= \left(\frac{R_p - R_f}{\sigma_p}\right)$ To evaluate risk adjusted return of investment.

R_p – Investment return
σ_p = Std. dev.
R_f = Risk free return.

$SR_x = \frac{0.14 - 0.4}{0.22} \approx 0.456$

$SR_y = \frac{0.09 - 0.04}{0.10} \approx 0.5$

$SR_z = \frac{0.20 - 0.04}{0.35} \approx 0.457$

Ranking $SR_y > SR_z > SR_x$.

Portfolio Z has return (absolute) of 20% and disproportionate risk of $\sigma_p = 35\%$

* c) A Y rational investor can leverage if ~~a~~ he borrows at $R_f = 4\%$ to match portfolio's (2) target risk of 35%. By scaling

$E[R_{lev.}] = R_f + \sigma_{target} \times SR_y$

$= 4\% + (35\% \times 0.5) \approx 21.5\%$. exceeds 20%.

8. a) GMV Portfolio – represents specific combination of risky assets that yields lowest possible variance of investments.

It matters because as max. risk reduction by adjusting weights of diff. assets based on individual correlation.

c) Capital Market Line (CML). is drawn by finding the line that starts at Rf on vertical axis and runs tangent to risky asset.

Slope $= \frac{E(R_m) - R_f}{\sigma_m}$ it represents Sharpe Ratio of market Portfolio